

AI Risk Prediction & Early Triage Assistant

Empowering Clinicians with Predictive Intelligence

TEAM

360 HORIZON (TETRA004)

INSTITUTION

NUV - ISEN Méditerranée

EVENT

TetraTHON 2026 - HealthTech Track

PRESENTERS

; Malak BOUDJELLAL

Ama Vivi EKOUVO

lyne-vanelle.maptue-fotso

The Healthcare Challenge

The Crisis

Lifestyle-related chronic conditions are experiencing a **severe surge** in global incidence.

Primary Bottleneck: Delayed diagnosis. Patients are often identified only after irreversible complications develop.

Target Pathologies: Diabetes, CAD, Stroke, and CKD.

Operational Gaps

-  **Restricted consultation windows** for primary care physicians.
-  **Lack of Clinical Decision-Support Systems (CDSS)** at the point of care.
-  **Fragmented access** to updated evidence-based triage guidelines.

Project Objective & Vision

"To engineer an Artificial Intelligence decision-support platform that empowers healthcare professionals to identify high-risk patients early."



Early Identification

Using routine, non-invasive clinical indicators to spot risks before they escalate.



Explainable AI

Providing actionable insights that clinicians can understand and trust.



Actionable Triage

Clear referral recommendations to optimize the patient care pathway.

Functional Scope & Features



Clinical Data Input

Intuitive entry form for patient baseline parameters, demographics, vital signs, and routine laboratory test results.



Multi-Pathology Scoring

Independent risk computation across four target pathologies: Diabetes, Coronary Artery Disease, Stroke, and Chronic Kidney Disease.



Explainable AI (XAI) Engine

Patient-level prediction breakdown identifying key risk drivers using **SHAP** (Shapley Additive exPlanations).



Urgent Triage Alerts

Automated visual notification system triggering **high-priority red alerts** for critical risk thresholds requiring immediate referral.

Data Pipeline & Machine Learning

CLINICAL DATASETS

Pathology	Dataset Name	Source	Records
Diabetes	Pima Indians Dataset	Public Medical Repository	768
Coronary Artery Disease	Framingham Heart Study	Framingham Cohort Study	4,240
Stroke	Stroke Prediction Dataset	Public Medical Repository	5,110
Chronic Kidney Disease	UCI CKD Dataset	Apollo Hospitals, India	400

XGBOOST ENGINE

- ✓ Superior performance on **tabular clinical data**.
- ✓ Seamless integration with **SHAP explainability**.

DATA STRATEGY

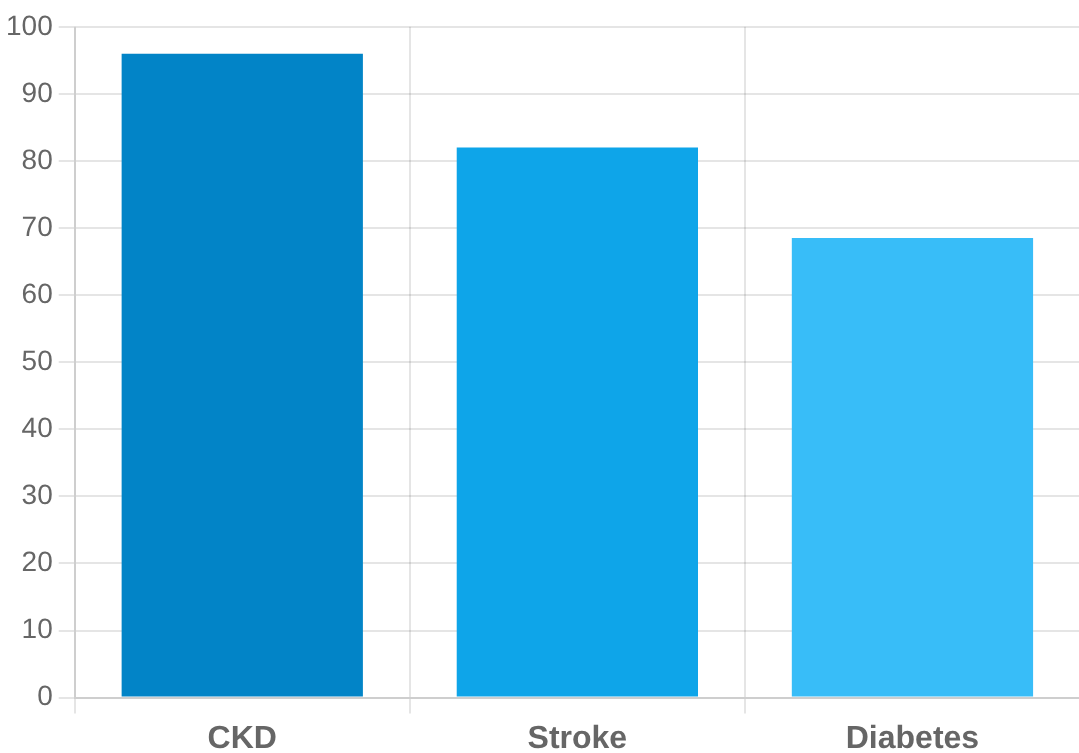
- 🛡️ **Missing Data:** Native handling of null values.
- ⚖️ **Class Imbalance:** Dynamic weight adjustments.

System Architecture & Tech Stack

Component		Technology Stack	System Role & Responsibility
	Frontend	React (Vite), Tailwind CSS	Dynamic clinical entry forms, risk display dashboard, and SHAP feature visualization.
	Backend API	FastAPI (Python 3.10+)	RESTful prediction endpoints, input validation, and ML pipeline orchestration.
	ML Models	XGBoost, Scikit-Learn	5 trained classifiers (4 disease models + 1 early CKD screening variant).
	Explainability	SHAP (TreeExplainer)	Real-time compute of feature attribution scores per prediction API call.

Experimental Results & Metrics

Model Recall (Sensitivity)



Performance Summary

Pathology	Recall	ROC-AUC
Chronic Kidney Disease	96.0%	0.99 (Acc)
Stroke	82.0%	0.28 (PR)
Diabetes	68.5%	0.83
Coronary Artery Disease	T-Dep	0.70-0.73

Key Insight: The CKD screening variant achieved near-perfect accuracy using basic clinical parameters, demonstrating high potential for early triage.

Building Clinical Trust

Explainable AI (XAI)

SHAP Attribution Graphs

Every prediction is accompanied by a localized breakdown of key risk drivers, mapping feature contributions to the final score.

Feature Transparency

Clinicians can see exactly which indicators (e.g., blood pressure, BMI) influenced the AI's decision.

Clinical Optimization

Risk Asymmetry Thresholds

Custom decision boundaries (e.g., 0.15 for Stroke) to maximize recall and minimize dangerous false negatives.

Practitioner Autonomy

The platform functions as a screening tool, preserving the final clinical judgment of the healthcare professional.

"Moving from 'Black-Box' algorithms to transparent, evidence-based clinical support."

Conclusion & Q&A

"Transforming reactive healthcare into proactive prevention through AI-driven intelligence."



Functional Prototype



Public Source Code



Technical Specification

Any Questions?

Team 360 HORIZON (TETRA004) | github.com/boudjellal-malak/TETRA004

